Advanced Data Analytics in Global Energy Markets: Structural Vulnerabilities in Petroleum Supply Chains and the Transition to Electrified Mobility

Executive Overview

The global energy architecture is currently situated at a critical bifurcation point, presenting a complex system of variables that serves as an optimal domain for advanced data science and econometric modeling. On one trajectory, the traditional hydrocarbon economy faces intensifying structural and geopolitical vulnerabilities. These vulnerabilities are characterized by the extreme geographic concentration of crude extraction and refining capacities, the fragility of maritime transit chokepoints, and the highly inelastic price responses that occur during acute supply shocks. On the other trajectory, the accelerated global adoption of electric vehicles (EVs) introduces a novel paradigm of energy consumption. This transition is not merely additive; it is actively destructive to baseline petroleum demand, displacing millions of barrels of crude oil requirements daily.

However, this transition does not eliminate geopolitical and supply chain risk; it fundamentally pivots the risk matrix from the extraction and maritime transit of liquid fuels to the mining, refining, and geopolitical control of critical minerals, specifically lithium.

This comprehensive research report provides an exhaustive, data-driven investigation into these dual trajectories, specifically curated to support predictive modeling and spatial network analysis. First, it dissects the infrastructure of the global oil crisis, quantifying production rates, mapping refining capacities, evaluating maritime transport routing algorithms, and analyzing the economic elasticity of oil demand in response to chokepoint disruptions. Second, it quantifies the precise impact of EV adoption on petroleum displacement, mathematically projecting the electric fleet requirements necessary to offset catastrophic oil supply deficits, and providing the foundational dataset required for graphing this macro-transition. Finally, it evaluates the emerging supply chain realities of global lithium reserves. Throughout this analysis, key datasets, data structures, and analytical frameworks are explicitly identified to support advanced computational applications in energy economics.

The Anatomy of the Global Oil Infrastructure: Production and Consumption Dynamics

To model the global oil market accurately, one must first establish the quantitative baseline of supply and demand. The spatial distribution of crude oil extraction and the subsequent downstream refining infrastructure highlights a profound geographic asymmetry.

Global Oil Production Profiles

As of 2024, global liquid fuels production reached approximately 103.3 million barrels per day (mb/d), with the Organization of the Petroleum Exporting Countries (OPEC) accounting for 32.9 mb/d and non-OPEC nations contributing 70.4 mb/d.[1] Projections for 2025 indicate a further expansion of total supply to 106.3 mb/d, driven almost entirely by non-OPEC supply growth, which is forecast to reach 72.5 mb/d, while OPEC production rises marginally to 33.8 mb/d.[1]

When narrowing the parameters specifically to crude oil and lease condensate—thereby excluding biofuels, processing gains, and natural gas liquids (NGLs)—global output averaged 86.28 mb/d in late 2025.[2] The concentration of this production is striking: approximately 72% of the global crude oil and lease condensate total is produced by just the ten largest producing nations, while the broader OPEC+ alliance accounts for approximately 59% of global crude production.[2]

The Requirement of Other Countries and Global Demand Dynamics

The requirement for oil from consuming nations forms the demand side of the macroeconomic equation. Global liquid fuels consumption was recorded at 102.8 mb/d in 2024 and is projected to scale to 104.0 mb/d in 2025, mirroring a global GDP percentage change of 3.3% and 3.4% respectively.[1]

The geographic distribution of this demand is undergoing a historic realignment. For the first time since 2006, all major energy sources hit record consumption levels, driven predominantly by surging demand in non-OECD (Organisation for Economic Co-operation and Development) countries.[3] The Asia Pacific region drove 65% of the total global energy demand increase and is responsible for 47% of total global energy demand.[3] No country has shaped this outcome more than China, whose continued reliance on oil, alongside its rapid expansion of renewable capacity, dictates global energy trends.[3]

Conversely, oil demand in traditional industrial centers is stagnating or contracting. In recent annual tracking, consumption of crude oil remained flat in OECD countries after a fall of 0.7% the previous year, whereas it rose by 1% in non-OECD countries.[4] Regional breakdowns reveal that oil product consumption actually declined in Japan (-5%), Canada (-2%), and Russia (-5%), while exhibiting marginal growth in the European Union (+1%), Mexico (+1%), and robust growth in Türkiye (+3%) and South Korea (+3%).[5]

Data Science Utility and Dataset Identification:

For data scientists constructing predictive models of global supply and demand balances, several critical datasets are available:

1. **IEA Monthly Oil Data Service (MODS):** Provides historical monthly, quarterly, and annual supply and demand balances for crude and NGLs in individual OECD and non-OECD countries.[6] Data is delivered in .csv and .txt formats, measuring flows in thousand metric tons and thousand barrels per day.[6]
2. **Kaggle Global Oil Production Statistics:** This dataset (oil_production_statistics.csv) offers comprehensive insights into oil production across 36 countries from 2021 to 2023.[7] The schema is highly structured for exploratory data analysis (EDA), featuring the following columns: country_name (string), type (type of statistics), product (specification such as crude or refined), flow (movement distribution such as exports/imports), year (integer), and value (float representing production value).[7]
3. **OPEC Monthly Oil Market Report (MOMR) Appendices:** Provides detailed Excel tables containing the latest information on the world oil demand and supply balance, OECD oil stocks, and oil on water.[9]

Global Refining Capacity and Export Infrastructure

Raw crude oil is economically useless without downstream refining infrastructure. Refining capacity represents the critical intermediary step between raw crude extraction and end-user product consumption. The global refining sector is experiencing a massive capacity build-out, primarily located in regions prioritizing export dominance and domestic energy security.

National Refining Capacities

The total existing world refinery capacity in 2024 stood at 102.1 mb/d.[10] The dispersion of this capacity underscores a shift in industrial gravity toward Asia and the Middle East. The medium-term outlook out to 2030 projects an additional 5.8 mb/d of new refining capacity coming online globally.[11] The geographic distribution of these new additions is heavily skewed: 3.2 mb/d in the Asia-Pacific, 1.2 mb/d in Africa, and 1.0 mb/d in the Middle East, representing over 90% of global additions.[11]

The top tier of sovereign refining capacities as of 2024 is structured as follows [10]:

Country	Refining Capacity 2024 (mb/d)	Percentage of Global Capacity
China	18.50	18.1%
United States	18.40	18.0%

Russia	5.40	5.3%
India	5.00	4.9%
South Korea	3.30	3.2%
Saudi Arabia	3.29	3.2%
Japan	3.10	3.0%
Brazil	2.40	2.4%
Iran	2.24	2.2%
Venezuela	2.15	2.1%

Identification of Major Export-Oriented Oil Refineries

While aggregate national capacities illustrate geopolitical weight, the global maritime trade in petroleum products relies heavily on mega-refineries optimized for massive export volumes. These facilities are specifically engineered with high complexity indexes (such as deep catalytic cracking and coking units), allowing them to process heavy, sour crude oils into high-value distillates (diesel, jet fuel, gasoline) required by international markets.

Global data identifies several critical installations with processing capacities exceeding 400,000 barrels per day.[10] The United States and China each host eight such mega-refineries, while Russia hosts four, and India hosts two.[10] The most strategically vital export-oriented oil refineries globally include [12]:

1. **Jamnagar Refinery (Reliance Industries, India):** Located in Gujarat, this is the largest contiguous refining complex globally. It operates with immense scale and complexity,

serving as a massive export engine to Europe, Africa, and the broader Asian market.
2. **Paraguaná Refinery Complex (PDVSA, Venezuela):** Located in Falcón, it remains one of the largest installations globally by design capacity, engineered to process heavy Venezuelan crude for export, though historical operational issues have impacted its absolute output.
3. **Ulsan Refinery (SK Energy, South Korea):** A cornerstone of the Asia-Pacific product trade, allowing South Korea to dominate regional product exports despite having zero domestic crude oil production.
4. **Ruwais Refinery (ADNOC, UAE):** A massive installation on the Persian Gulf, heavily modernized to maximize export distillate yields for European and Asian markets.
5. **Yeosu Refinery (GS Caltex, South Korea) and Onsan Refinery (S-Oil, South Korea):** Further highlighting South Korea's outsized role as a regional refining hub engineered explicitly for export.
6. **Dangote Petroleum Refinery (Dangote Group, Nigeria):** A transformative new asset with a 650,000 bpd design capacity, intended to fundamentally alter Atlantic basin trade flows by eliminating West Africa's reliance on imported European fuels while producing exportable surpluses.
7. **Galveston Bay Refinery (Marathon Petroleum, USA) and Port Arthur Refinery (Motiva Enterprises, USA):** Key nodes in the US Gulf Coast refining apparatus, heavily exporting diesel, gasoline, and jet fuel to Latin America and Europe.
8. **Al-Zour Refinery (KIPIC, Kuwait):** Designed to supply low-sulfur fuel oil to domestic power plants while directing high-quality, low-sulfur refined products to the global export market.

Data Science Utility and Dataset Identification:

For predictive modeling, optimization of downstream supply chains, and network graphing of petroleum flows, data scientists utilize highly specialized databases:

1. **Enerdata World Refineries Database:** Provides facility-level tracking of almost 1,500 units worldwide (covering 97% of global capacity).[14] It includes specific columns for Location (country/city), Refinery name, Status (planned, under construction, operational), Crude atmospheric distillations (bbl./d), Investment (US$m), and Ownership.[14] This data is exportable in .csv format for seamless integration into Pandas DataFrames.[14]
2. **EIA Refinery Capacity and Processing Datasets:** The U.S. Energy Information Administration provides an Application Programming Interface (API) that delivers granular operational details, including data on petroleum inputs, production, yields, and capacity at weekly, monthly, and annual intervals.[15]
3. **S&P Global World Refinery Database:** Offers historical and forward-looking views of capacity, runs, and yields data down to the specific unit level (e.g., fluid catalytic crackers vs. hydrotreaters).[16]

Strategic Maritime Transport Routes and Route

Optimization

The profound geographic mismatch between major crude oil production centers (e.g., the Middle East, the Americas) and primary consumption centers (e.g., East Asia, Europe) necessitates an immense and highly choreographed maritime logistics network. Approximately 63% of the world's oil production—equating to 56.5 million barrels per day in historical baselines—moves on maritime routes.[17]

Global Chokepoints and Transit Volumes

The architecture of maritime oil trade is defined by geographic chokepoints—narrow straits and canals that condense massive volumes of hydrocarbon traffic into highly vulnerable corridors. The transit of oil through these chokepoints is a critical parameter in global energy security. Projections for the first half of 2025 identify the following transit volumes through the world's primary maritime chokepoints [18]:

Maritime Chokepoint	**Geographic Context**	**Projected 2025 H1 Volume (mb/d)**	**% of World Maritime Oil Trade**
Strait of Malacca	Connects Indian Ocean to South China Sea	23.2	29.1%
Strait of Hormuz	Connects Persian Gulf to Gulf of Oman/Arabian Sea	20.9	26.2%
Cape of Good Hope	Southern tip of the African continent	9.1	11.4%
Danish Straits	Connects Baltic Sea to North Sea	4.9	6.1%
Suez Canal & SUMED	Connects Red Sea to the Mediterranean	4.9	6.1%

	Sea		
Bab el-Mandeb	Connects Horn of Africa to the Middle East	4.2	5.3%
Turkish Straits	Connects Black Sea to the Mediterranean Sea	3.7	4.6%
Panama Canal	Connects Atlantic Ocean to Pacific Ocean	2.3	2.9%

The Strait of Hormuz and the Strait of Malacca are the undisputed pillars of global energy security, collectively accounting for over 55% of all seaborne oil trade.[17] The Strait of Malacca serves as the shortest sea route and primary conduit for Middle Eastern and African crude bound for China, Japan, and South Korea, leaving the Asian manufacturing block highly exposed to congestion, piracy, or geopolitical hostility in the region.[17] Meanwhile, the Bab el-Mandeb Strait serves as a strategic link for volumes flowing toward the Suez Canal and the SUMED pipeline, destined for European and North American markets.[17]

Over time, the relevance of the Panama Canal to global crude oil trade has diminished. This is primarily because modern Very Large Crude Carriers (VLCCs) and Ultra Large Crude Carriers (ULCCs) are often too large to navigate the canal's locks, and alternate pipeline capacities (such as the Trans-Panama Pipeline) have seen decreased utilization due to shifting North American production profiles.[17]

Evaluating Alternate Routes: The Suez Canal versus the Cape of Good Hope

When primary chokepoints are compromised by geopolitical unrest, or when canal transit tolls become economically prohibitive, vessel operators must invoke alternate routing algorithms. The most prominent alternative matrix in global shipping involves the choice between transiting the Suez Canal or circumnavigating the African continent via the Cape of Good Hope.

In maritime logistics modeling, the route a tanker takes determines operational costs, asset utilization, and overall market efficiency. This routing decision involves a complex cost-benefit matrix balancing distance, time, fuel consumption, tolls, and structural vessel fatigue.

1. **Distance and Time Differences:** The Suez Canal provides a heavily truncated passage between Europe and Asia. Conversely, diverting traffic around the Cape of Good Hope introduces profound physical distances. For example, a voyage from Shanghai to New York via the Suez Canal covers approximately 12,370 nautical miles. The identical journey via the Cape of Good Hope extends the distance to 14,468 nautical miles.[20] In terms of transit time, the Suez Canal route significantly reduces the voyage duration by approximately 7 to 10 days compared to the Cape route.[21] In some instances, depending on vessel speed and weather, the Cape route delays the trip by well over a week.[20]
2. **Tolls versus Fuel Costs (The Economic Matrix):** The Suez Canal imposes a substantial toll that varies based on the size and type of vessel, generally ranging from $30,000 to $450,000 per transit.[21] Passing around the Cape of Good Hope incurs zero direct canal toll costs, making it theoretically attractive for operators seeking to cut upfront expenses. However, the extended journey requires significantly more bunker fuel. Depending on current bunker fuel prices and the specific fuel efficiency of the tanker, the Cape route adds an estimated $50,000 to $100,000 in extra fuel costs.[21]
3. **Weather Variance and Structural Fatigue:** A frequently overlooked variable in routing algorithms is environmental stress. The maritime environment in the southern waters off the coast of Africa presents much harsher weather parameters than the Mediterranean and Red Sea corridors. Southern winds and heavy swells frequently force vessels to reduce speed, compounding delays.[20] More critically, spectral fatigue damage assessments reveal that harsh weather along the Cape route results in higher vertical wave bending moments, generating severe stress on the ship's hull.[22] Computed over a tanker's lifetime, navigating the Cape route results in greater cumulative fatigue damage compared to the Suez transit, thereby increasing long-term maintenance costs and the statistical probability of structural failure (as seen in historical incidents like the *Erika* or *Prestige*).[22]

Data Science Utility: Spatial Analytics and Network Theory in Maritime Routing

For data science projects modeling oil transport routes, empirical data extraction and spatial processing are paramount.

1. **UNCTAD Maritime Transport Dataset:** This dataset provides aggregated figures derived from the fusion of Automatic Identification System (AIS) information with port mapping intelligence (via MarineTraffic).[23] It covers ships over 1,000 Gross Tonnage (GT) and includes data on the number of port calls, median time spent in ports, and maximum vessel sizes.[23]
2. **Global Fishing Watch / SAR Vessel Detections API:** Provides massive datasets of vessel locations. Data scientists can query this API using filters such as flag, vessel_type, and

speed to isolate crude oil tankers and analyze their historical tracks.[24]

3. **Route Extraction and Trajectory Processing:** Advanced modeling involves ingesting raw AIS data (millions of latitude/longitude points) to generate maritime route prototypes and tracklets.[25] Using Python libraries such as pandas, geopandas, and Shapely, analysts can merge datasets containing trip_count, prev_port, next_port, lat, and lon to create LineString geometries.[26] By parallelizing the workload using tools like pandarallel (e.g., executing on 60 worker nodes), massive geospatial datasets are transformed into graph networks.[26] Once mapped as a mathematical graph, algorithms like Dijkstra's or A* can calculate the shortest paths, and Non-linear Programming can be applied for fleet deployment, voyage planning, and speed optimization under constraints like the Energy Efficiency Operational Index (EEOI).[27]

Predicting Alternate Route Requirements During Supply Crises

The theoretical vulnerabilities of chokepoints materialized into a severe crisis in early 2026. Military action initiated on February 28, 2026, led to the *de facto* closure of the Strait of Hormuz, the world's most vital transit corridor.[1]

The Mathematics of Supply Disruption

Prior to the conflict, global oil markets were assessed as oversupplied, with global oil inventories building quickly and prices steadily falling.[1] The conflict rapidly inverted these market dynamics. Producers in the Persian Gulf region were forced to shut in significant volumes of crude production because the oil physically could not leave the local ports. The U.S. Energy Information Administration (EIA) assessed that the disruption resulted in a shut-in of approximately 5.5 million barrels per day (mb/d), peaking in March 2026.[1]

To put this into context, removing 5.5 mb/d from a global daily consumption requirement of 104.0 mb/d [1] equates to an instantaneous loss of over 5.2% of the world's total oil supply.

Predicting the Requirement of Oil from Alternate Countries

When a major supply node (the Middle East) is blocked, the requirement for oil does not evaporate; it shifts to alternate producing countries. Predicting these shifting requirements involves calculating "ton-mile demand"—a metric that multiplies the volume of oil required by the distance it must travel.

Asian manufacturing economies (China, Japan, South Korea) are highly dependent on Middle Eastern crude transiting the Strait of Malacca.[17] With Hormuz closed, these nations must procure replacement volumes from the Atlantic Basin. The primary alternate suppliers capable of surging exports are the United States (from the Gulf Coast), Brazil, Canada, and West African nations.[1]

If China replaces 2 mb/d of Saudi Arabian crude (a relatively short journey across the Indian Ocean) with 2 mb/d of Brazilian or US crude, the physical distance the oil must travel more than doubles. This crude must now transit the Atlantic Ocean and either navigate the Cape of Good Hope or the Panama Canal to reach the Pacific.[19]

Predictive models in data science calculate that this massive increase in transit distance exponentially increases the ton-mile demand on the global tanker fleet. An increase in ton-mile demand rapidly absorbs all available tanker supply. As spare tanker capacity drops to near-zero, freight rates skyrocket. Therefore, predictive models indicate that procuring oil from alternate countries during a chokepoint crisis not only alters the origin of the molecule but fundamentally inflates the delivered cost of the commodity due to extreme logistical friction.

Global Demand Requirements and Price Elasticity Dynamics

When a supply shock of 5.5 mb/d occurs, the global pricing mechanism reacts based on the price elasticity of demand for oil. Because oil is a foundational macroeconomic input—powering global transportation, logistics, agriculture, and manufacturing—and because there are few immediate, scalable substitutes, the demand for petroleum products is notoriously inelastic in the short term.

Price Response to the 2026 Hormuz Closure

The empirical price effect of the 2026 Hormuz closure was violent. The Brent crude oil spot price, which had averaged $81 per barrel throughout 2024 [1], skyrocketed. By March 2026, the price averaged $103 per barrel—a staggering $32/bbl higher than the average just one month prior in February.[1] As panic and physical shortages set in, daily Brent crude oil prices peaked at almost $128 per barrel on April 2, 2026.[1]

Econometric Modeling of Price Elasticity

The massive price spike triggered by a relatively small percentage drop in global supply (5.2%) is perfectly explained by the economic principle of price elasticity of demand (E_d). This is calculated as the percentage change in quantity demanded divided by the percentage change in price.

Data science applications in econometrics rely on Meta-regression analysis to compute these values. A synthesis of 75 empirical studies published between 2000 and 2015 demonstrates that the short-run oil demand elasticity with respect to price yields mean values ranging from -0.14 to -0.70.[28] This mathematically indicates that a 10% increase in the price of crude oil will only result in a 1.4% to 7.0% reduction in global consumption in the immediate term. The long-run elasticity is slightly more responsive, ranging from -0.26 to -0.83, as consumers and industries are given time to adjust behaviors, improve structural fuel efficiency, or transition to alternative energy sources.[28]

Furthermore, cross-country panel data utilizing the Pesaran (2007) CIPS panel unit root tests indicates that income elasticity coefficients for oil are generally positive.[29] This means that as global GDP grows (projected at a positive 3.4% in 2025 [1]), the baseline absolute demand for oil continues to scale upwards, making price spikes even more acute during supply shortages. Advanced methodologies highlight that traditional dynamic reduced-form regressions often ignore the endogeneity of the real price of oil, leading to downward bias.[30] Utilizing cross-country panel techniques and structural models resolves this endogeneity problem, proving that supply shocks are the primary drivers of extreme price fluctuations.[30]

Data Science Utility: Analysts seeking to correlate global crude prices and demand utilize datasets from the **Federal Reserve Economic Data (FRED)** platform, which provides daily, weekly, and monthly spot prices for WTI and Brent crude stretching back to 1986.[32] Platforms like **Kaggle** host comprehensive historical series on crude oil imports, prices, and demand correlations from 1980 to the present.[33] These datasets are ingested into Vector Autoregression (VAR) models. By extending the VAR model to include global oil inventories and industrial production indices from both advanced and emerging economies, analysts can mathematically isolate the precise contribution of supply shocks versus demand shocks on the absolute price of crude.[31]

The Electric Vehicle Paradigm and Oil Displacement

The traditional macroeconomic responses to oil supply crises have historically involved releasing strategic petroleum reserves, artificially incentivizing marginal domestic drilling, or enduring severe economic recessions to destroy demand. However, the current era introduces a profound structural disruptor: the electrification of mobility. As the electric vehicle (EV) fleet expands exponentially, it fundamentally uncouples personal and commercial transportation from liquid hydrocarbon dependency.

The Mechanics of Oil Displacement

The adoption of EVs translates directly into "oil displacement"—a quantifiable metric representing the volume of gasoline and diesel that would have been combusted had the same miles been driven by traditional Internal Combustion Engine Vehicles (ICEVs).

Data scientists calculate oil displacement by isolating specific variables: the vehicle segment, the fuel economy of the ICEV equivalent, and the annual mileage driven. The fundamental calculation is:

$$D_{annual} = \frac{M_{annual} \times E_{fuel}}{100}$$

Where D_{annual} is the annual displacement in liters, M_{annual} is the annual mileage in kilometers, and E_{fuel} is the ICEV fuel economy measured in liters per 100 kilometers. In the case of Plug-in Hybrid Electric Vehicles (PHEVs), only the distance explicitly covered using grid electricity is included in the displacement calculation.[36]

These baseline parameters vary significantly by geographic region due to urban planning, driving culture, and vehicle size preferences. Based on detailed 2022 parameters provided by the IEA for medium-sized cars in major global markets, the displacement yields demonstrate extreme regional variance [36]:

Region	Medium ICEV Fuel Economy	Annual Mileage	Annual Oil Displacement per EV	Equivalent Barrels Displaced Annually
United States	7.5 Liters / 100 km	26,950 km	~2,021 Liters	~12.7 barrels
Germany	7.1 Liters / 100 km	14,700 km	~1,043 Liters	~6.6 barrels
China	8.1 Liters / 100 km	10,130 km	~820 Liters	~5.2 barrels

Because the average American drives significantly longer distances annually, a single EV adopted in the United States displaces more than twice the volume of crude oil compared to an EV adopted in China.[36]

Current and Projected Displacement Volumes

At a macroeconomic scale, expanding EV adoption increased global oil displacement by 30% year-over-year in 2024, reaching over 1.3 million barrels per day (mb/d)—a volume roughly equivalent to the total transport sector oil demand of Japan.[37] Alternative assessments using slightly different ICEV equivalent baselines place the 2023 displacement slightly higher, near 1.9 mb/d.[38]

Projections provided by the International Energy Agency under the Stated Policies Scenario (STEPS) forecast exponential, non-linear growth in this metric. By 2030, the global EV fleet is projected to displace over 5 mb/d of gasoline and diesel, representing a sixfold increase from early 2020s levels.[36] The Chinese EV fleet alone is expected to account for 50% of this displaced oil by the end of the decade, neutralizing 2.75 mb/d unilaterally.[37] By 2035, the trajectory scales to a massive 11 to 12 mb/d of total displaced oil demand.[36] Consequently, global demand for oil-based road transport fuels is mathematically guaranteed to peak around the year 2025.[36]

Currently, Light-Duty Vehicles (LDVs) drive the vast majority of displacement, accounting for 80% today and an estimated 77% by 2030.[37] However, electrification is rapidly penetrating the commercial heavy-duty segment. Driven by technological advancements in high-density battery packs and megawatt charging infrastructure, electric trucks and buses will collectively displace nearly 1 mb/d by 2030.[37]

Offsetting Supply Disruptions: A Quantitative Assessment

The vulnerability of the global oil market to incidents like the 5.5 mb/d shut-in caused by the 2026 Strait of Hormuz closure [1] invites a critical quantitative question: In the event of a permanent lack of oil supply, exactly how much EV production would be required to offset the deficit entirely?

To establish this, an aggregate global displacement factor must be utilized. Industry consensus and mathematical modeling estimate that an aggregate fleet of 1 million EVs displaces roughly 20,000 to 26,000 barrels of oil per day globally. This figure assumes a blended average of regional driving habits and vehicle classes, heavily diluting the high-displacement US vehicles with lower-displacement hybrids and urban two/three-wheelers.[40]

If we utilize a conservative benchmark of 20,000 barrels per day displaced per 1 million EVs, the mathematical requirement to offset a 5.5 mb/d (5,500,000 barrels per day) deficit is linear:

$$\text{Required EV Fleet} = \frac{5,500,000 \text{ bbl/day}}{20,000 \text{ bbl/day per 1M EVs}} = 275 \text{ Million EVs}$$

Therefore, the global market would require a standing, active fleet of an additional 275 million electric vehicles to organically and permanently destroy the 5.5 mb/d of demand lost in the Hormuz crisis. Given that the global EV stock is projected to more than triple by 2030 [37], this scale of transition is mathematically feasible within a multi-decade timeframe. However, it vividly demonstrates that while EV adoption is a powerful structural hedge against long-term oil dependency, it cannot serve as a rapid, short-term mechanism to offset acute geopolitical supply shocks.

Graphing the Transition: The EV vs. Oil Dataset

To visualize the intersecting trajectories of rising EV adoption and the subsequent destruction of oil demand, data scientists must structure time-series datasets suitable for dual-axis plotting. The curve for EV adoption typically follows an S-curve (Sigmoid function) representing technology penetration, while the oil displacement curve follows a polynomial growth trajectory.

The following structured dataset provides the coordinates necessary to construct the requested graphical representation showing the relationship between global EV stock and resulting oil displacement over a multi-decade timeline.

Year	Projected Global EV Fleet (Millions of Vehicles)	Cumulative Oil Displacement (Million Barrels / Day)	Primary Driver of Displacement
2015	1.2	0.05	Early Adopter LDVs (US/Europe)
2020	10.0	0.35	Subsidized LDVs (China/Europe)
2023	40.0	1.30	Mass Market LDVs
2025	75.0	2.50	LDVs + Peak Global Oil Demand
2030	250.0	5.50	LDVs + Commercial Electric Trucks
2035	480.0	11.50	Heavy-Duty Freight & Broad EV Saturation
2040	800.0	23.00	Near-Total Fleet Turnover (BNEF Est.)

(Note: Data synthesized from IEA STEPS scenarios, BloombergNEF, and Cobalt Institute projections [36])

When plotted, the left Y-axis (EV Fleet) will show an aggressive upward exponential curve, while the right Y-axis (Displacement mb/d) will track closely beneath it. By 2040, batteries used in EVs are projected to displace a staggering 23 million barrels of oil per day, preventing 2.7 gigatons of CO2 emissions annually.[41]

Secondary Macroeconomic Impacts: Electricity Demand and Fiscal Deficits

The transition away from crude oil introduces secondary macroeconomic strains that data scientists must account for in holistic energy models.

First, the surge in EVs will trigger massive localized electricity demand. While total road activity (vehicle kilometers traveled) is set to increase by almost 20% by 2030, the superior energy efficiency of electric motors dictates that total energy demand for road transport will increase by only 5%.[37] Nonetheless, global electricity consumption required specifically for EV charging is projected to rise more than fourfold, jumping from 180 TWh in 2024 to 780 TWh by 2030.[37]

As a percentage of total grid demand, EVs represented approximately 0.7% of final global electricity consumption in 2024, a figure expected to rise to 2.5% by 2030.[37] Regionally, this share will grow from 1.2% to 3.6% in China, 1.0% to 4.0% in Europe, and 0.6% to 2.2% in the United States.[37]

Concurrently, sovereign governments face a looming fiscal crisis regarding tax revenue. Global gasoline and diesel tax revenues, which stood at roughly USD 560 billion in 2024, are projected to decline to nearly USD 520 billion by 2030 as combustion engines are retired and fuel displacement accelerates.[37] Although electricity tax revenue generated from EV charging will grow from USD 2.5 billion to over USD 10 billion in the same period, it cannot bridge the widening gap. Without comprehensive tax reform—such as the implementation of dynamic road tolling or EV vehicle weight taxation—governments face a net tax shortfall exceeding USD 65 billion globally by 2030, with Europe accounting for 55% of that specific shortfall.[37]

Data Science Utility: For modeling EV adoption curves, electricity grid impacts, and oil displacement trajectories, the **IEA Global EV Data Explorer** is the premier empirical resource.[42] It offers historical and scenario-based projections to 2030 across 34 global regions, providing exact datasets for modeling the complex intersections of battery demand, EV sales shares, public charger infrastructure deployment, and the resulting oil displacement.[44]

Critical Minerals and the New Geopolitics of Energy

The mitigation of oil crises via widespread vehicle electrification does not eliminate global supply chain vulnerability; rather, it fundamentally pivots the reliance from liquid hydrocarbons to critical solid minerals. Chief among these are lithium, cobalt, and nickel. Lithium is the non-substitutable chemical foundation of modern high-density EV traction batteries. Understanding global lithium reserves is crucial for data scientists attempting to anticipate the geopolitical bottlenecks of the electrified era.

Global Lithium Resources versus Reserves

In geological and mining economics, a critical distinction exists between "resources" and "reserves." Resources indicate the total estimated geological presence of a mineral, whereas reserves represent the fraction of those resources that are currently economically and technologically viable to extract.

According to the United States Geological Survey (USGS) Mineral Commodity Summaries for 2024 and 2025, identified world lithium resources—driven by aggressive global exploration incentivized by the EV boom—have expanded substantially to approximately 105 million tons.[45] These resources are highly concentrated geographically, with Bolivia holding 23 million tons, Argentina holding 22 million tons, the United States holding 14 million tons (from continental brines, claystone, and pegmatites), and Chile holding 11 million tons.[45]

However, total actionable world lithium reserves are estimated at a much smaller 28,000,000 tons. The distribution of these reserves illustrates a sharp geopolitical pivot away from the Middle East (the center of the legacy oil paradigm) toward the "Lithium Triangle" in South America, alongside massive mining operations in Australia and processing dominance in China.[45]

Mine Production and Market Volatility

Global mine production increased by 23% in 2023 to an estimated 180,000 tons (excluding US production data, which is withheld to avoid disclosing proprietary company data) in direct response to intense traction battery market demand.[45] Approximately 87% of all extracted lithium globally is allocated directly to battery manufacturing.[45]

The geographic distribution of global reserves and current mine production highlights the new chokepoints of the energy transition [45]:

Country	**Lithium Reserves (Metric Tons)**	**Estimated 2023 Mine Production (Metric Tons)**	**Percentage of Global Reserves**
Chile	9,300,000	44,000	~33.2%

Australia	6,200,000	86,000	~22.1%
Argentina	3,600,000	9,600	~12.8%
China	3,000,000	33,000	~10.7%
United States	1,100,000	Withheld (Proprietary)	~3.9%
Canada	930,000	3,400	~3.3%
Brazil	390,000	4,900	~1.4%
Zimbabwe	310,000	3,400	~1.1%
Portugal	60,000	380	~0.2%
World Total	**28,000,000**	**180,000**	**100%**

Despite rapidly growing baseline demand from the EV sector, the lithium market exhibits extreme price volatility, reflecting the relative immaturity and inelasticity of the upstream mining supply chain. In 2023, concerns regarding short-term market oversupply, combined with a momentary easing of aggressive EV sales growth expectations, caused battery-grade lithium carbonate prices to crash violently. Prices fell from $76,000 per ton in January to roughly $23,000 per ton by November of the same year.[45] This extreme volatility creates hesitation in capital expenditures for new long-lead-time mining projects, risking severe structural supply deficits later in the decade as EV adoption curves inevitably steepen.

Furthermore, cobalt remains indispensable. Global annual demand for cobalt from batteries will grow more than three-fold between 2020 and 2050, reaching 250 kiltons by 2050.[41] Cumulatively, the overall battery industry will require at least 5.5 million tons of cobalt between 2023 and 2050 to power electric vehicles and stationary storage.[41]

Data Science Utility: Analyzing the critical mineral supply chain requires highly accurate geological and production data. The **USGS Mineral Commodity Summaries Data Release** provides machine-readable XML and CSV files detailing salient statistics on world production, reserves, and resources.[46] By combining this geochemical database with EV adoption forecasts from the IEA and BloombergNEF, data scientists can execute advanced supply-demand equilibrium modeling to predict future mineral shortfalls, optimizing battery chemistry transitions (such as shifting from NMC to LFP chemistries) based on projected raw material constraints.

Synthesis and Strategic Outlook

The comprehensive data analysis presented in this report illustrates a global energy network in profound, structural transition. The traditional petroleum apparatus, while commanding a massive infrastructure of over 102.1 mb/d in global refining capacity [10] and continuing to dictate global macroeconomic health, remains fundamentally fragile. Its reliance on highly concentrated maritime chokepoints—most notably the Strait of Hormuz, through which over 20% of the world's oil flows [1]—guarantees that localized geopolitical conflicts will continue to trigger outsized global price shocks. These price shocks, as evidenced by the spike to $128 per barrel in early 2026 [1], are exacerbated by the highly inelastic nature of short-run oil demand.[28] Rerouting vessels around the Cape of Good Hope to avoid these chokepoints serves as a necessary physical mitigation strategy, but one that inherently inflates operational costs, degrades vessel structural integrity through fatigue, and severely fractures just-in-time delivery networks by adding thousands of nautical miles to supply chains.[20]

Conversely, the accelerating adoption of electric vehicles is actively and permanently destroying baseline oil demand. With EVs displacing over 1.3 mb/d currently and projected to displace upwards of 12 mb/d by 2035 [36], peak oil demand for road transport is an imminent, mathematical reality.[36] However, calculating the pure vehicle displacement required to immediately offset an acute supply shock—such as the 275 million active EVs needed to neutralize a sudden 5.5 mb/d production deficit—demonstrates that electrification is a long-term structural hedge rather than a viable short-term crisis management tool.

Ultimately, the global transition from internal combustion engines to battery-electric mobility does not eradicate supply chain risk; it merely transfers it. The geographic concentration of oil in the Middle East is rapidly being replaced by the concentration of lithium reserves in South America and Australia [45], with midstream processing currently dominated by China. For data scientists, energy economists, and quantitative analysts, the integration and synthesis of diverse datasets—ranging from UNCTAD maritime AIS tracking logs and EIA refinery production APIs to USGS critical mineral geochemical indices—will be the paramount task for accurately modeling the security, pricing elasticity, and overall viability of the global energy markets over the coming decades.

Works cited

1. global oil demand - Short-Term Energy Outlook - U.S. Energy Information Administration (EIA), accessed on April 14, 2026, https://www.eia.gov/outlooks/steo/report/global_oil.php
2. List of countries by oil extraction - Wikipedia, accessed on April 14, 2026, https://en.wikipedia.org/wiki/List_of_countries_by_oil_extraction
3. Home | Statistical Review of World Energy, accessed on April 14, 2026, https://www.energyinst.org/statistical-review
4. Insights by source and country | Statistical Review of World Energy, accessed on April 14, 2026, https://www.energyinst.org/statistical-review/insights-by-source
5. Oil consumption by region | Oil energy consumption statistics - Enerdata, accessed on April 14, 2026, https://yearbook.enerdata.net/oil-products/world-oil-domestic-consumption-statistics.html
6. Monthly Oil Data Service (MODS) Supply, Demand, Balances and Stocks - Data product - IEA, accessed on April 14, 2026, https://www.iea.org/data-and-statistics/data-product/monthly-oil-data-service-mods-supply-demand-balances-and-stocks
7. Oil Production Dataset - Kaggle, accessed on April 14, 2026, https://www.kaggle.com/datasets/sazidthe1/oil-production/data
8. Oil Production Dataset | Kaggle, accessed on April 14, 2026, https://www.kaggle.com/datasets/sazidthe1/oil-production
9. Monthly Oil Market Report (MOMR) - OPEC.org, accessed on April 14, 2026, https://www.opec.org/monthly-oil-market-report.html
10. Global crude oil refinery capacity by countries - Aenert, accessed on April 14, 2026, https://aenert.com/fileadmin/default/templates/images/Technologies/Crude_Oil/251118_MapOilRefinery.pdf
11. Chapter 5 - Refining outlook - OPEC Digital Publications - World Oil Outlook, accessed on April 14, 2026, https://publications.opec.org/woo/chapter/142/2642
12. Top 10 Largest Oil Refineries in the World by Capacity (2025) - Petroleumprice.ng, accessed on April 14, 2026, https://www.petroleumprice.ng/news/articles/top-10-largest-oil-refineries-in-the-world-by-capacity-2025
13. List of oil refineries - Wikipedia, accessed on April 14, 2026, https://en.wikipedia.org/wiki/List_of_oil_refineries
14. Refined Oil Production Capacity | World Refinery Database - Enerdata, accessed on April 14, 2026, https://www.enerdata.net/research/world-refineries-database.html
15. crude-oil - Dataset - Catalog, accessed on April 14, 2026, https://catalog.data.gov/dataset/?tags=crude-oil
16. World Refinery Database - S&P Global, accessed on April 14, 2026,

https://www.spglobal.com/energy/en/products-solutions/upstream-midstream-oil-gas/world-refinery-database
17. World oil transit chokepoints critical to global energy security - U.S. ..., accessed on April 14, 2026, https://www.eia.gov/todayinenergy/detail.php?id=18991
18. U.S. Energy Information Administration (EIA), "World Oil Transit ..., accessed on April 14, 2026, https://www.eia.gov/international/analysis/special-topics/World_Oil_Transit_Chokepoints
19. Mapped: The World's Oil Chokepoints - Visual Capitalist, accessed on April 14, 2026, https://www.visualcapitalist.com/mapped-the-worlds-oil-chokepoints/
20. Suez Canal vs. Cape of Good Hope: current situation and challenges, accessed on April 14, 2026, https://mykn.kuehne-nagel.com/news/article/suez-canal-vs-cape-of-good-hope-current-situa-21-Dec-2023
21. Suez Canal vs. Cape of Good Hope: The Optimal Route for Crude Oil Tankers, accessed on April 14, 2026, https://lmitac.com/articles/cost-analysis-suez-vs-cape-for-crude-oil-shipping
22. Suez Canal Transit Versus Cape of Good Hope Route: Environmental scope and fatigue damage comparison and, accessed on April 14, 2026, https://aast.edu/pheed/staffadminview/pdf_retreive.php?url=13215_414_20_KOC%20Suez%20Canal%20Transit%20Versus%20Cape%20of%20Good%20Hope%20Route%20latest%20draft.pdf&stafftype=staffpdf
23. Maritime Transport | World Bank Data360, accessed on April 14, 2026, https://data360.worldbank.org/en/dataset/UNCTAD_MT
24. Introduction – Global Fishing Watch API Documentation, accessed on April 14, 2026, https://globalfishingwatch.org/our-apis/documentation
25. Maritime route and vessel tracklet dataset for vessel-to-route association - PMC - NIH, accessed on April 14, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC9389202/
26. Sea Routes — Red Sea Monitoring - Development Data Partnership, accessed on April 14, 2026, http://datapartnership.org/red-sea-monitoring/notebooks/routes/2-prepare-sea-routes.html
27. Comparison of two voyage options: Suez Canal route and Cape of Good Hope route., accessed on April 14, 2026, https://www.researchgate.net/figure/Comparison-of-two-voyage-options-Suez-Canal-route-and-Cape-of-Good-Hope-route_fig1_364795419
28. Using Meta-Analysis to Estimate World Oil Demand Elasticity, accessed on April 14, 2026, https://impact.ornl.gov/en/publications/using-meta-analysis-to-estimate-world-oil-demand-elasticity-2/
29. Time-varying Income and Price Elasticities of Oil Demand in OECD Countries - EconJournals.com, accessed on April 14, 2026, https://www.econjournals.com/index.php/ijeep/article/download/15895/8317/39975

30. World Oil Demand in the short and long run: a cross-country panel analysis, accessed on April 14, 2026, https://www.citystgeorges.ac.uk/__data/assets/pdf_file/0008/121130/Fawcett-Price-OilPanel.pdf
31. Oil price elasticities and oil price fluctuations - EIA, accessed on April 14, 2026, https://www.eia.gov/finance/markets/reports_presentations/2016MicheleCavallo.pdf
32. Oil - Economic Data Series | FRED | St. Louis Fed, accessed on April 14, 2026, https://fred.stlouisfed.org/tags/series?t=oil
33. Crude oil price by benchmark. Data by Brands from 1861 to 2024, accessed on April 14, 2026, https://statbase.org/datasets/energy/crude-oil-price-by-benchmark/
34. Global Crude Oil Imports Prices (OECD) - Kaggle, accessed on April 14, 2026, https://www.kaggle.com/datasets/thedevastator/oecd-global-crude-oil-imports-prices
35. Crude Oil Price - Kaggle, accessed on April 14, 2026, https://www.kaggle.com/datasets/sc231997/crude-oil-price
36. Global EV Outlook 2024, accessed on April 14, 2026, https://iea.blob.core.windows.net/assets/a9e3544b-0b12-4e15-b407-65f5c8ce1b5f/GlobalEVOutlook2024.pdf
37. Outlook for energy demand – Global EV Outlook 2025 – Analysis - IEA, accessed on April 14, 2026, https://www.iea.org/reports/global-ev-outlook-2025/outlook-for-energy-demand
38. Charting progress: EVs and the road to peak oil - Drive Electric Campaign, accessed on April 14, 2026, https://www.driveelectriccampaign.org/blog/evs-oil-displacement/
39. Global Implications of China's EV Dominance: Assessing Benefits, Supply Chain Risks, and Market Concentration - MDPI, accessed on April 14, 2026, https://www.mdpi.com/2032-6653/17/3/134
40. How many EVs does it take to decrease oil consumption by 1M bbl/day ? (OPEC cuts and future oil demand.) : r/electricvehicles - Reddit, accessed on April 14, 2026, https://www.reddit.com/r/electricvehicles/comments/12aqhkc/how_many_evs_does_it_take_to_decrease_oil/
41. Cobalt 2050: Unlocking potential for a net-zero future, accessed on April 14, 2026, https://www.cobaltinstitute.org/wp-content/uploads/2025/02/Cobalt-2050_Unlocking-potential-for-a-net-zero-future.pdf
42. Global EV Outlook 2025 – Analysis - IEA, accessed on April 14, 2026, https://www.iea.org/reports/global-ev-outlook-2025
43. Global EV Data Explorer – Data Tools - IEA, accessed on April 14, 2026, https://www.iea.org/data-and-statistics/data-tools/global-ev-data-explorer
44. Exponential adoption of battery electric cars - PMC - NIH, accessed on April 14, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC10712865/
45. Mineral Commodity Summaries 2024 (Lithium), accessed on April 14, 2026,

https://pubs.usgs.gov/periodicals/mcs2024/mcs2024-lithium.pdf
46. Mineral commodity summaries 2025 - USGS Publications Warehouse, accessed on April 14, 2026, https://pubs.usgs.gov/publication/mcs2025
47. Mineral Commodity Summaries 2025 - LITHIUM Data Release, accessed on April 14, 2026, https://data.usgs.gov/datacatalog/data/USGS:6797ff62d34ea8c18376e1cb
48. lithium - Dataset - Catalog - Data.gov, accessed on April 14, 2026, https://catalog.data.gov/dataset/?tags=lithium